Trenitalia
ETR1000 VZI-6

Windscreen Wiper

Revision	Author	Date	Revision Description
00.1	R. Schiavo	27/09/2023	Draft Release Generated from DOORS module /Platform.-.Zefiro-Europe/..//2F_02.02.-.Wash Wipe-VF.-.Zefiro-Europe/Wash Wipe-VF.CONCEPT, baseline 1.3 Doc 3EST000225-7900_A
00	R. Schiavo	27/09/2023	First Emission

Content

1 Introduction 4

2 Train Configuration 4

3 Definitions and abbreviations 4

4 System Overview 5

5 System Configuration 5

6 Functional requirements 6

7 Activation of Windscreen Wiper + Washer 6

8 Function Design 6

13 Requirements 6

17 Signal Interface 8

18 I/O Interface 8

19 Supplier system 8

20 TCMS SW 8

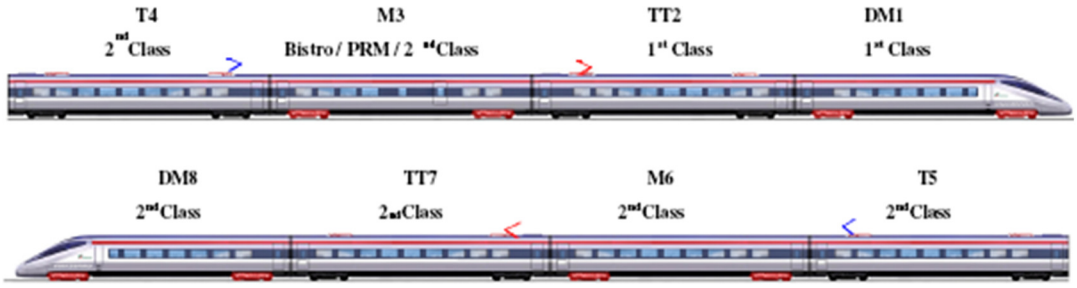
21 Bus Interface 8

1 Introduction

This document is a function specification for the control of the *windscreen wiper* system for ETR1000 / VZI6 for customer Trenitalia.

2 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

3 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
TSI	Technical specification of interoperability

4 System Overview

5 System Configuration

The task of the windscreen wiper and washer is to clean a specific field of the windscreen according to TSI and standards specifications. The washing system is integrated in the wiper. The complete system consists of a pneumatic motor to move the wiper, a pneumatic pump for the washer, a tank, a control unit and a control panel on the drivers desk. It is controlled under normal conditions via electrical signals from 2 switches on the drivers desk and in case of an electrical failure by a pneumatic emergency button. It is possible to chose between different speeds of wiping. The complete system is considered to be not safety relevant.

6 Functional requirements

7 Activation of Windscreen Wiper + Washer

8 Function Design

9 *Normal condition*

The driver can chose to activate the wiper by using a switch on the drivers desk. It is possible to chose the speed of the wiper in two modes.
Interval: 1 double stroke of the wiper every specific periode of time
Continuously: a specific number of strokes per second
It is possible to use the washing system by pressing a button. The washer will be active as long as the button is pressed. During the washing the wiper moves with a specific speed.

10 *Failure cases*

1. Wiping speed selected but no action of the system due to electrical failure.
2. Wiping speed selected but no action of the system due to pneumatic and possible additional failure.
3. Washing selected but no water is hoisted on the windscreen
4. Washing selected but wiper is not active

11 *Failure detection*

1. Detected by the driver.
2. Detected by the driver.
3. Detected by the driver.
4. Detected by the driver.

12 *Failure handling*

1. By using the pneumatic emergency button the field of the failure can be narrowed down to the electrical system. Train has to be repaired during corrective maintenance. Service still possible according to visibility situation. (to be included in drivers manual)
2. Train has to be repaired during corrective maintenance. Service still possible according to visibility situation.
3. Train has to be repaired. Service still possible according to visibility situation. Amount of water has to be checked during maintenance and refilled according to maintenance plan. No use of washing system is allowed, when no water is hoisted on the screen due to risk of damage of the waterpump. (to be included in drivers manual)
4. By using the pneumatic emergency button the field of the failure can be narrowed down to the electrical system. Service still possible according to visibility situation. Train has to be repaired. (to be included in drivers manual)

13 Requirements

14 *Washer button*

ID	2F_02.02.Zefiro-Europe.TRS.12
Description	There shall be a control on the drivers desk and when operated fluid shall come over the windscreen.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.02.Zefiro-Europe.CONCEPT.65	A push button shall be integrated on the drivers desk. The button shall have an unstable position. (the device does not remain in the position after its application, it moves automatically in the former stable position)	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_02.02.Zefiro-Europe.CONCEPT.48	The push button shall be wired to the windscreen wiper control unit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.02.Zefiro-Europe.CONCEPT.66	If the button is pushed, the water pump shall be activated	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.67	If the button is pushed, the wiper shall start to operate with 1 wipe/second	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.68	If the push button is released, the wiping shall stop after 5 s	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A

2F_02.02.Zefiro-Europe.CONCEPT.75	If the button is released, the water pump shall stop operating	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A

15 *Wiper speed selection switch*

ID	2F_02.02.Zefiro-Europe.TRS.11
Description	The washer and wiper functions shall be operated with a control that have intervall operation and continuosly operation modes.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.02.Zefiro-Europe.CONCEPT.90	The wash wiper system shall be powered up when the respective cab is activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.02.Zefiro-Europe.CONCEPT.59	A switch with 6 positions shall be installed on the drivers desk. The switch shall have stable positions. (the device remains in the position after its application)	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_02.02.Zefiro-Europe.CONCEPT.46	The switch shall be wired to the windscreen wiper control unit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.02.Zefiro-Europe.CONCEPT.69	1. Position Windscreen wiper shall move to/be in park position	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.70	2. Position Windscreen wiper shall operate with 5 wipes per min	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.71	3. Position Windscreen wiper shall operate with 10 wipes per min	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.72	4. Position Windscreen wiper shall operate with 20 wipes per min	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.73	5. Position Windscreen wiper shall operate with 35 wipes per min	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A
2F_02.02.Zefiro-Europe.CONCEPT.74	6. Position Windscreen wiper shall operate with 55 wipes per min	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A

16 *Wiper emergency switch*

ID	2F_02.02.Zefiro-Europe.TRS.22
Description	To increase availability of the wiping system there shall be an emergency button in the driver's desk.
Safety level	N/A

To increase the availability of the wiping system, a pneumatic switch shall be integrated in the drivers desk. It shall be directly connected via pipes to the wiper motor.

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.02.Zefiro-Europe.CONCEPT.87	The pneumatic emergency button shall break the power supply of the wiper control unit when triggered.	Derived Requirement	4S_02.01.04.-.Windscreen Wiper and Washing Systems		N/A

17 Signal Interface

The system shall provided by the supplierincluding control unit and switches. Therefore every necessary Signal is internal and will not be descibed in this SFRS

18 I/O Interface

19 Supplier system

Input

Nr	Functional name	Type	Linked to requirement
2F_02.02.Zefiro-Europe.CONCEPT.62	n/a		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.02.Zefiro-Europe.CONCEPT.63	n/a		

20 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.02.Zefiro-Europe.CONCEPT.30	n/a		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.02.Zefiro-Europe.CONCEPT.31	n/a		

21 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_02.02.Zefiro-Europe.CONCEPT.32	n/a		

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_02.02.Zefiro-Europe.CONCEPT.33	n/a		